

Meri Kahani: A Gamified Solution to Teach Computational Thinking to Female Teenagers in Low Resource Communities

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ABSTRACT

This paper describes the design of a mobile based gaming application - Meri Kahani - created to teach computational thinking skills to school going teenagers in underdeveloped areas of Pakistan. We explore the use of gamification to teach computational thinking through level-based learning in a Pakistani context. This paper's final design demonstrates how gamified learning, rewarding techniques, and feminine themes can be used to attract female teenagers towards computational thinking. This paper also discusses the evaluation and usability testing results conducted on 16 school-going female teenagers. We hope that through this study, we have taken the first step towards nurturing an interest in young females for computational thinking and overcoming the gender gap that adversely affects female involvement in Computer Science in Pakistan.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Accessibility; Accessibility technologies; Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

Computational Thinking, Gamification, Teenage Girls, Low-Socioeconomic, Storytelling, Pakistan

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1 INTRODUCTION

In recent years, various countries that have undertaken curriculum redesign for primary and secondary education have endorsed computational thinking as an essential mindset for teenagers. Computational thinking helps build and enhance the ability to deal with complex scenarios, open-ended problems and communicate and work with others to achieve a common goal [1]. However, in a third-world country like Pakistan, millions of children are deprived of top-quality education, most of whom belong to impoverished backgrounds. Young females belonging to these low socio-economic households are amongst the worst affected. In 2018, it was reported that 32% of primary-school-age girls were out of school in Pakistan, and only 13% of girls were still in school by ninth grade [2]. Attitudes regarding girls' education vary significantly across the country. Implicit social biases and stereotypical cultural norms paired with lower quality of education markedly curtail the number of opportunities available for them to enhance their computational thinking skills and develop practical problem-solving abilities, something which can eventually help them play a more profound role in Pakistan's growth. As a result, a severe gender gap exists within the computer science field in Pakistan.

In this research work, we discuss the conception and design of a gamified and educational approach that focuses on honing computational thinking skills in young females and motivating and empowering them to pursue STEM and Computer Science fields. Through a user study conducted on 22 young female students, we identified our proposed application's core functionalities and the specific user needs that it is meant to cater. Using the insights gathered, we proposed a solution in the form of a high-fidelity prototype. In the following sections, we document the stages we underwent to formulate the final high-fidelity design of 'Meri Kahani' and preliminary evaluation results collected through usability testing.

2 RELATED WORK

Computational Thinking (CT) is a problem-solving method that involves conceptualizing, designing systems, and developing

abstractions in which individuals relate the problems with logical thinking [21]. After its rise in the twenty-first century, many countries incorporated it into the early years of their curriculum. Past research has shown that developing students' CT abilities has many positive impacts. It can improve their coding performance [3]. While learning different CT skills, they become more efficient problem solvers equipped enough to face various problems [4]. An individuals' understanding of CT can be measured based on the number of steps they have defined to solve the problem. To teach students CT, educational systems have taken upon different methods like traditional out-of-classroom activities, team-based problem solving, and different game-based solutions (board games etc.). However, developing countries like Pakistan have failed to adopt it into their educational system, putting the nation at a significant disadvantage. Recent research has shown that educational programs do not explicitly train teachers to teach CT to students. Due to this, many individuals graduate from high school clueless about what CT is and how it relates to problem-solving and algorithmic thinking [5]. An in-depth review of Pakistan's primary educational curriculum (the matriculation system) showed that the current syllabus is usually limited to teaching students about computers in general and lacks emphasis on CT [6]. It is built upon rote memorization, predicting question patterns, and outright cheating [17]. This shows how current methods of teaching CT in schools are ineffective.

Moving on, there is a dire need to break the stereotype of teenage girls pursuing the computer science field where gender disparity has become an issue. These gender differences arise from cultural beliefs, gender role-socialization, lack of confidence in females, and how computers are taught and practiced in schools [16]. Recent surveys of women in tech in Pakistan have shown that females are losing interest in Computer Science from grade school to higher education [7]. The current number of females dropping out of the IT field in academia and the industry has been increasing, which needs to be changed [7]. A significant reason behind the declining interest is the prevalent mindset in Pakistan that females are best suited for medical professions [20]. Another reason is that there is a lack of sources of inspiration for females. Research has shown that women look up to female role models for inspiration and before making tough decisions [18]. To encourage female participation in the IT field, research has shown to start teaching them CT skills at the early age of 13-16, which is critical for learning [8].

Furthermore, gamification, as a method of learning, has gotten increased attention over the past years. Out of 60 studies investigated on Game-Based Learning (GBL), 22 (32%) studies resulted in better performance of the students as these studies defined the learning objectives precisely and targeted the specific learning content, contrary to 38 (56%) studies which measured the effectiveness of games on learning achievement [9]. It allows people to directly teach domain-specific thinking skills and metacognitive skills, improving learning and performance [10]. Research on gamification pedagogy has also proven its advantages in allowing less advanced students to benefit from the learning activities and providing complex learning environments appropriate for the acquisition of learning with meaningful materials that encourage the construction of relevant knowledge using the suitable models [22]. Moreover, past research has also shown that gender differences

play a significant role in a Digital Game-Based Learning (DGBL) approach when it comes to affecting learning outcomes. An online survey for a study showed that female respondents were less attracted to games with sexual gender role stereotyping [23]. This shows that games designed for females need to incorporate a more 'feminine' theme. However, a game-design workshop to promote young girls' interest in computing showed improvements in their attitudes towards computers and their self-reported confidence and competence with computers [24]. Joint research conducted by the Department of Innovative Technologies in Education and Department of Computing at the National University of Sciences and Technology in Islamabad, Pakistan, found that girls demonstrated significant positive body language in most sessions using a GBL (Game-Based Learning) application [9]. Similarly, girls were more focused and confident in most of the app instructional sessions [9]. For girls in Pakistan, gamification is also a good option because it allows them to learn at their own pace at home using smartphones if their guardians do not allow them to go out of their homes for education or pursue a relevant field.

Currently, many different applications and websites exist that are trying to teach CT skills effectively. Some female-specific applications are Smartgurlz, Luna and Cat, and Nancy Drew, whereas some gender-neutral applications are ScratchJr, CodingGalaxy, and Grasshopper. However, most of these applications have drawbacks when taken into the local Pakistani context. None of them have personally relatable themes meaningful for Pakistani teenage girls because of the culture and language differences. According to numerous studies, females are intrigued by personally relevant content [11]. Another significant drawback of most of these applications is that they do not support online collaboration. Researchers say cooperation is exceptionally beneficial for female adolescents [12]. Lastly, we found a lack of motivation, cheering, and confidence-building within the applications. Previous research has shown that females' confidence and performance improve after acquiring positive feedback [13]. They are shown to gain faith and rely on their own reasoning when motivated to think [14].

With respect to Pakistan, there has been no significant work to focus on developing a healthy environment where CT could thrive, especially for Pakistani teenage girls. Existing applications and websites are inaccessible for numerous underprivileged Pakistani adolescent girls since they are mostly English and lack the local cultural context [15]. The curriculum has also not been updated to focus on improving CT abilities rather than general knowledge about computers themselves. Our research aimed to work on the syllabus by introducing up-to-date topics that are more tailored to today's technology era. We wanted to achieve this by making the concepts easier to understand and immersing the students in digital game-based learning methods as tools in and out of the classroom. Moreover, our research kept in mind the gender disparity and intended to motivate and encourage females to pursue STEM fields, especially computer science, by fostering CT.

3 USER RESEARCH

We conducted user research to understand the gender gap in CT within Pakistan and assess school-going female students' current CT skills. Furthermore, we studied their use of technology and its

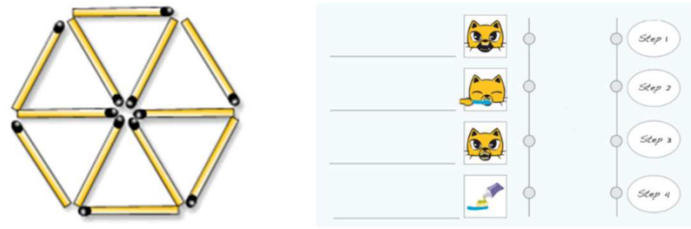


Figure 1: The images above show the unplugged activities that our participants performed

impact on their daily life. Members of our research group comprised 22 school-going girls aged 13-17 years, belonging to Pakistan's developing areas with limited access to the internet. We conducted user research in two community schools of Lahore – Allied School and Government District Girls High School.

Our research method consisted of a mixed-method approach focusing on the following techniques: (i) contextual inquiry, (ii) semi-structured interviews, and (iii) user observation. The semi-structured interviews were held in an isolated environment to minimize distraction and allow the participants to focus better. The interview questions aimed to gauge the participants' current interaction with technology and their interest in gamified learning techniques. To conduct the contextual inquiry and test the CT skills of the participants, we designed three unplugged problem-solving activities based on various CT concepts for the participants to attempt: (i) Diagram Labelling: Identifying the correct order of steps for brushing teeth, (ii) Arranging the Matchsticks: Make the maximum number of triangles from 12 matchsticks, and (iii) The Paper Plane problem: Identify the correct order of steps to make a paper plane. These activities were carefully chosen after studying the CT curriculums for primary schools abroad. They aimed at testing creative problem solving, decomposition, algorithms, algorithm design, and logical reasoning concepts of the participants. Figures 1 and 2 show images of the activities. Furthermore, we made the participants play "Mazes and More," a labyrinth-solving mobile game, to observe their comfort level with mobile games. The participants' behavior, emotions, and facial expressions were observed while performing these activities, including the activity completion time, completion rate, and the frequency of assistance provided to them.

Significant findings from our user research include the following:

Gender Gap and Lack of Awareness: Computer Science in Pakistan has an increasing gender gap that can be attributed to gender stereotypes and a lack of awareness regarding the importance of CT skills. 45% of participants showed little to no interest in Computer Science. A computer was nothing more than just a "device" for them. During the interview, one of the participants stated, "My parents want me to become a doctor as it is a good profession for girls." [P5] Females from an early age are encouraged to play with feminine toys such as dolls, whereas boys are encouraged to play with devices such as video gaming. As a result, the perception that computers are masculine devices becomes ingrained within young girls' minds.

Poor Computational Thinking Skills: Schools in Pakistan focus more on following rigid book curriculums instead of applied learning techniques.

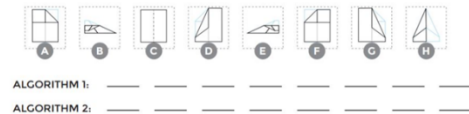


Figure 2: The image above shows another unplugged activity performed by our participants.

54.5% of our participants were Computer Science students; however, their knowledge was only limited to the curriculum taught at school, highlighting the flaw of rote learning in our education system. Only one participant who "learned about computer science from her uncle during her free time" had exposure to knowledge beyond that taught in school. While the participants performed unplugged activities, we deduced that only 9.0% of the participants successfully solved task 1 and task 2. As for task 3, only one participant was able to solve it completely. 36.4% of the participants frequently asked for hints. These observations suggest that most participants had a weak understanding of CT concepts and found the activities challenging.

Current Use of Technology: Females living in the underdeveloped areas of Pakistan have restricted access to technology. 73% of our participants did not own a personal device, and many of them used their mother's mobile phone whenever the need arose. However, it was surprising that all 22 participants had access to an internet connection at home. While the participants played Mazes and More, it was observed that 80% of them were able to navigate through the application without any assistance. All the participants completed the game level successfully. The maximum time taken was 35 seconds, and the minimum time was 9.5 seconds, whereas the average was 17.6 seconds. These observations suggest that the participants had smooth interactions with technology and were comfortable using it.

Digital Game-Based Learning: Gamified learning is a concept foreign to the education system of Pakistan. However, 86.3% of the participants stated that they would prefer learning concepts through games rather than books. This claim is further supported by observations made while the participants played the Mazes and More game. There was a 100% completion rate for the mobile game. The participants took more time to complete unplugged activities and scored low. Whereas, the participants completed Mazes and More in approximately 17.6 seconds, significantly lower than the time taken to complete unplugged activities. The participants used words such as "easy, enjoyable, and fun" to describe their experience with the game. One of the participants claimed, "Games are more

interesting and easier to learn through." [P 7] Moreover, 77.3% of the participants claimed that they played games such as "Subway Surfer and Temple Run" on their phones, indicating their interest in gamification.

Language Accessibility: One unusual finding of our study was that most participants were comfortable with the English language. Since our target audience stems from the lesser-developed segments of Pakistan, where education is primarily Urdu-instructed, we had initially anticipated that most of our users would feel more at home while using an application in Urdu. However, contrary to our anticipation, 73% of the participants preferred that the application proposed should be developed in English language. 13.6% of the participants preferred Urdu language only, whereas the remaining 13.4% suggested using English and Urdu in the application to make it more accessible.

Inspiration from Female Role Models: Our study revealed that most women look up to female role models for inspiration and before making tough decisions. Due to the massive dearth of females in Pakistan's technological field, there are very few leading females that our girls can take inspiration from. Out of the 22 people we interviewed, only 1 of them was genuinely passionate about Computer Science. The rest were generally uninspired. One of the participants stated that "there is a lack of females within the field of computer science. However, the one person I look up to is Arfa Karim [a Pakistani female computer prodigy who became the youngest Microsoft Certified Professional (MCP) in 2004]." [19] [P 18] These findings support the importance of highlighting existing women's achievements in computer science to serve as an inspiration for young girls.

4 DESIGN AND IMPLEMENTATION

After evaluating our user research through extensive brainstorming and paper prototyping, we concluded that a mobile application called "Meri Kahani" would best fulfill our requirements through gamified learning. In order to attract more girls towards the application, a feminine theme is used in our design. Our user research and literature show that a feminine theme makes the application more fun, engaging, and personally relatable for female teenagers. Our findings also illustrate that most participants preferred female characters and role models because it is more inspiring and relatable for them. The feminine theme is implemented in our application by adding female avatars, female role models, female instructors, and a female color palette. Our application also includes features that allow the users to purchase avatar items and customize them to their liking to make their gameplay experience more enjoyable. Furthermore, our solution focuses majorly on teaching within a Pakistani context. This is achieved through a Pakistani-themed map interface, using illustrations of well-known local monuments and vehicles such as Chinchy. The Pakistani-themed interface and the feminine aspect make our solution unique, as no mobile application in Pakistan currently teaches CT in such a relatable manner and offers such a fascinating learning experience for young girls.

To effectively teach CT, we designed a curriculum involving the most important concepts such as pattern recognition, abstraction, and sequence recognition. The curriculum was designed after interviewing female teenagers at local schools, going through local

computer science textbooks, and identifying the loopholes in them. Our prototype teaches these essential concepts through plugged activities and gamification. The girls we interviewed showed a greater interest in a more interactive gamified solution focusing on storytelling. A female avatar explores the Pakistani map within the game, which includes different game levels, focusing on teaching a specific CT concept. For further help, a video lesson is available on the map screen that teaches the CT concept tested on the upcoming game level. Female instructors teach the concepts in the video lessons specifically to inspire young girls. As the game progresses, the levels and concepts being taught increase in difficulty. Hints and visual cues are also provided in each game level to offer support and make the gameplay experience convenient for users. To increase the accessibility of our solution, game levels can be played offline, but the users must sign up and log in if they wish to save and track their progress. To further improve accessibility, the solution provides bilingual support, offering both English and Urdu language as the medium of instruction. To encourage and motivate the users through a sense of achievement, they are rewarded with coins upon each game level's successful completion. In addition to this, daily coin rewards are awarded to each user to motivate them to use the application regularly. Another significant theme that our solution revolves around is inspiring young girls by highlighting the achievements of female role models in computer science. This is achieved through the inspiration section within our prototype, which consists of blogs and videos highlighting Pakistani female role models' achievements in technology.

We initially tested our solution using low-fi paper prototypes. Based on the results and user feedback, the final hi-fi prototype was designed on Figma. We implemented four game levels within our final prototype that focus on crucial CT concepts that increase in difficulty as the game progresses. We hope to ensure that our solution can successfully improve CT concepts in young girls through a fun gamified learning experience.

5 EVALUATION AND RESULTS

We conducted usability tests in a detailed and meticulous manner to assess whether our prototype fulfills its objectives. To gauge our digital interface's usability from our target audience's perspective, we circulated a form on social media to get school-going girls to sign up. Due to the pandemic, since the only way of reaching out to the users was through social media, we could not contact many people from underdeveloped areas. Therefore, most of our participants during testing belonged to English-Medium schools from developed areas of the country. There were a total of 15 participants out of whom 40% were interested in the field of Computer Science, and 53.3% stated as being "possibly interested". The participants ranged from grade 8 to 12. The testing procedure was carried out remotely on Zoom, a videoconferencing software, allowing us to record the whole testing procedure. The test procedure was supervised by two members and was divided into three phases. Firstly, the consent form was shared with each participant, asking their permission to record the test, i.e., their audio and screen. A basic overview of the whole project and usability testing goals and purpose were explained to the participants. The participants were then asked to interact with the shared prototype and complete the set of tasks



Figure (a)

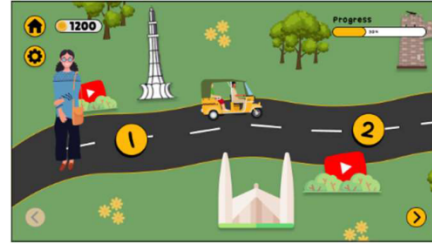


Figure (b)

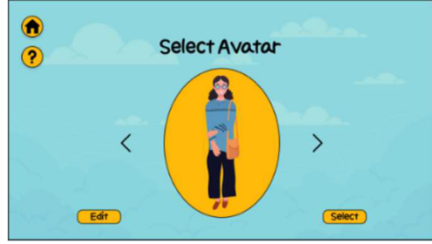


Figure (c)



Figure (d)

Figure 3: Figure (a) shows the main home screen of our game in the Urdu version. Figure (b) shows the game map. Figure (c) shows the screen where users can select an avatar of their own choice. Figure (d) shows the first level of the game – a jigsaw puzzle

provided. Finally, a feedback/usability scale form was shared with the participants to gauge their overall experience.

Four different methods were employed to evaluate our measures: direct observations, analysis of the recorded session, direct questions with the interviewees, and the feedback/usability scale form. Our usability testing focused on quantifying the performance usability metrics and the preference measures of our users. Quantitatively, on average, participants completed 85% of the tasks, taking 9 seconds to complete a task. The users committed an average number of 8 errors per task and took an average time of 8.3 seconds to recover from a task. Moreover, an average of 1.67 requests for help were made to us by users while performing the tasks. The post-testing feedback form showed that 73% of the participants believed they would play this game frequently, not a single one of them finding it complex to understand. 80% of the participants felt that this game could teach them necessary CT skills.

Qualitatively, users expressed their liking for the game prototype's overall concept and the Pakistani context that it was set in. One of the participants expressed, "I love the idea of an application teaching CT." Most of them found the design and the feminine interface very appealing. "The Pakistani clothes are so cute" was another typical quote we received from the participants. When asked about the game's difficulty level, most of them replied with "You guys really put some thought into making this app easy for us!". Moreover, they loved the Urdu language feature of the prototype. "Who would have thought of a CT application in our native language!" one of them exclaimed. However, the participants also found some shortcomings in our design. Some feature placements like the logout icon and video icon did not match the users' mental model, resulting in their memory failing to assist them in some of

the tasks. Some users also complained of unclear game instructions causing ambiguity in their minds. For example, a participant worriedly asked, "How do I start the application and begin playing the game?".

The findings we managed to extract from this phase made us realize a few shortcomings to our prototype that need to be improved before the application enters its development phase. By the end of usability testing, most of our participants were satisfied and convinced with our design's innovative nature. However, we still require future studies to determine the efficacy and success of Meri Kahani.

6 CONCLUSION AND FUTURE WORK

Our research aimed to improve female teenagers' CT skills in Pakistan, which has become a crucial part of the modern-day school curricula in developed countries. The sustenance of computational thinking competencies through academics is understood not to be limited to a particular gender. In Pakistan, however, little emphasis is given to inducing these competencies amongst teenage girls. Our user research also reflected this lack of focus on CT skills in Pakistan's general school curriculum. Our solution took a gamified learning approach and developed a high-fidelity prototype that focused on enhancing and teaching CT skills to female teenagers in Pakistan. After extensive usability testing and evaluation, we came across a few shortcomings in our application's design. However, we found our solution to be generally effective at conveying the relevant CT concepts to our target users. It was effective because our application provided bilingual support, employed a feminine theme, embedded personalized avatars, and made the game environment and map more culturally relatable to induce utmost engagement

from the users' part. Future work on this application will aim to refine our high-fidelity prototype's design based on our evaluation results. After the prototype is developed into a fully functional mobile game, Meri Kahani has significant potential to be a useful learning application in and out of the classroom. It will be a fuller and scalable application with the same content framework. Meri Kahani can be easily adaptable in different countries with its theme, artwork, illustrations, and language adjusted according to the cultural context. Ensuring compatibility for local Pakistani languages would also make the application culturally appropriate on a provincial level in Pakistan. Moreover, future work on our study will comprise detailed research and analyses of a broad range of CT skills that can be embedded into our application. We hope to see Meri Kahani evolve with newer and improved game levels, avatars, and inspirational material to play an integral role in teaching CT skills and decreasing gender disparity by motivating female teenagers to pursue the STEM fields as a career.

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